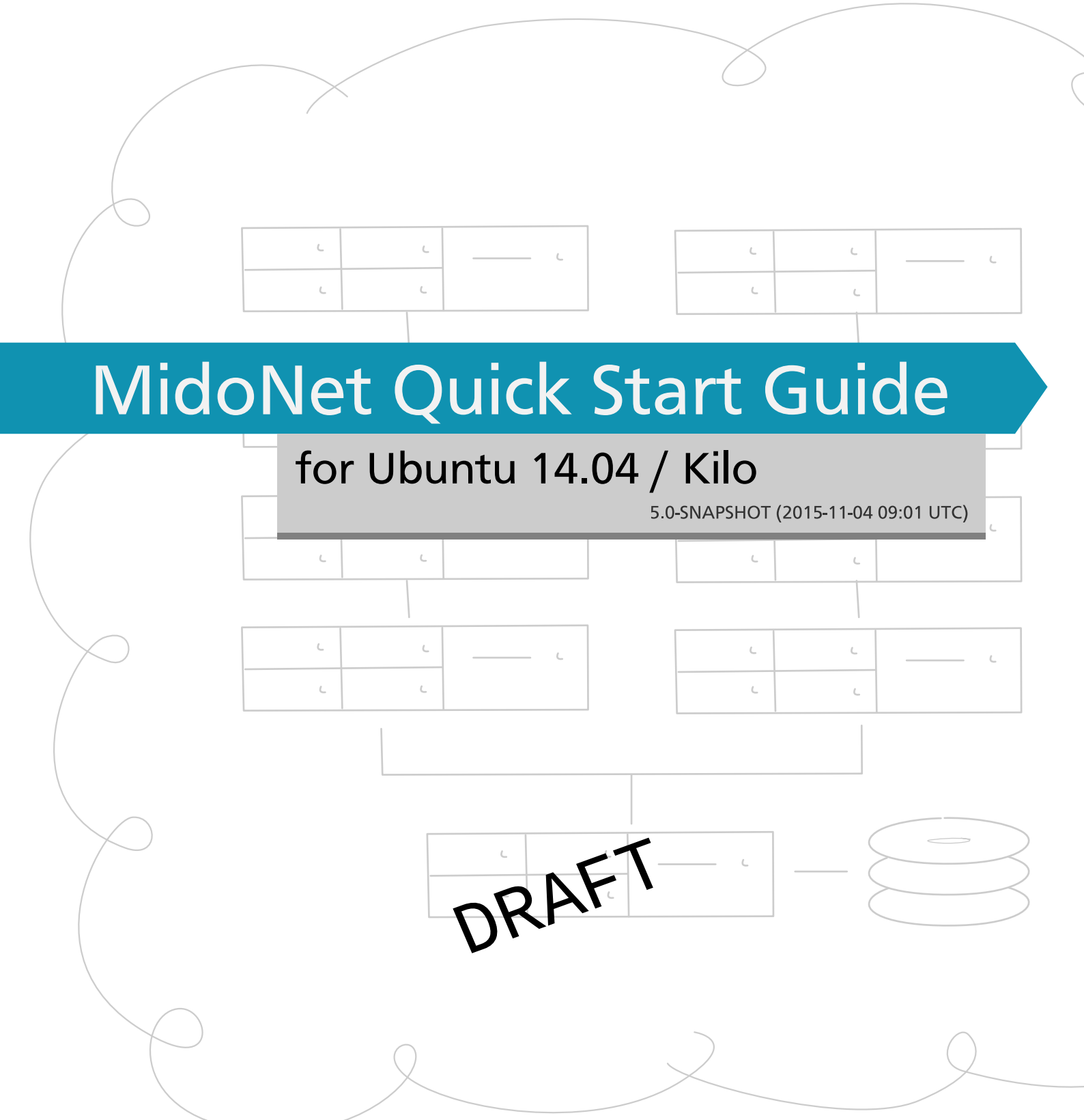
MidoNet Quick Start Guide

for Ubuntu 14.04 / Kilo

5.0-SNAPSHOT (2015-11-04 09:01 UTC)



midonet

docs.midonet.org

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MidoNet Quick Start Guide for Ubuntu 14.04 / Kilo

5.0-SNAPSHOT (2015-11-04 09:01 UTC)
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MidoNet is a network virtualization software for Infrastructure-as-a-Service (IaaS) clouds.

It decouples your IaaS cloud from your network hardware, creating an intelligent software abstraction layer between your end hosts and your physical network.

This guide walks through the minimum installation and configuration steps neccessary to use MidoNet with OpenStack.

Caution

This document is a DRAFT. It may be MISSING relevant information or contain UNTESTED information. Use it at your own risk.

Note

Please consult the MidoNet Mailing Lists or Chat if you need assistance.

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Preface

Conventions

The MidoNet documentation uses several typesetting conventions.

Notices

Notices take these forms:

Note

A handy tip or reminder.

Important

Something you must be aware of before proceeding.

Warning

Critical information about the risk of data loss or security issues.

Command prompts

$ prompt

Any user, including the root user, can run commands that are prefixed with the $ prompt.

 # prompt

The root user must run commands that are prefixed with the # prompt. You can also prefix these commands with the **sudo** command, if available, to run them.

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1. Architecture

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Hosts and Services 2

This guide assumes the following example system architecture.

OpenStack Controller Node:

- Controller Node (**controller**)

Compute Node:

- Compute Node (**compute1**)

Since MidoNet is a distributed system, it does not have the concept of a Network Node as being used with the default OpenStack networking plugin. Instead it uses two or more Gateway Nodes that utilize Quagga to provide connectivity to external networks via the Border Gateway Protocol (BGP).

- Gateway Node 1 (**gateway1**)
- Gateway Node 2 (**gateway2**)

Three or more hosts are being used for the MidoNet Network State Database (NSDB) cluster which utilizes ZooKeeper and Cassandra to store virtual network topology and connection state information:

- NSDB Node 1 (**nsdb1**)
- NSDB Node 2 (**nsdb2**)
- NSDB Node 3 (**nsdb3**)

Important

Ideally, both the ZooKeeper transaction log and Cassandra data files need their own dedicated disks, with additional disks for other services on the host. However, for small POCs and small deployments, it is ok to share the Cassandra disk with other services and just leave the ZooKeeper transaction log on its own.

The *MidoNet Agent (Midolman)* has to be installed on all nodes where traffic enters or leaves the virtual topology. In this guide this are the **controller**, **gateway1**, **gateway2** and **compute1** hosts.

The *Midonet API* can be installed on a separate host, but this guide assumes it to be installed on the **controller** host.

The *Midonet Command Line Interface (CLI)* can be installed on any host that has connectivity to the MidoNet API. This guide assumes it to be installed on the **controller** host.

The *Midonet Neutron Plugin* replaces the ML2 Plugin and has to be installed on the **controller**.

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Hosts and Services

Controller Node (controller)

- General
  - Database (MariaDB)
  - Message Broker (RabbitMQ)
- OpenStack
  - Identity Service (Keystone)
  - Image Service (Glance)
  - Compute (Nova)
  - Networking (Neutron)
    - Neutron Server
    - DHCP Agent
    - Metadata Agent
  - Dashboard (Horizon)
- MidoNet
  - API
  - CLI
  - Neutron Plugin

Compute Node (compute1)

- OpenStack
  - Compute (Nova)
  - Networking (Neutron)
- MidoNet
  - Agent (Midolman)

NSDB Nodes (nsdb1, nsdb2, nsdb3)

- Network State Database (NSDB)
  - Network Topology (ZooKeeper)
  - Network State Information (Cassandra)

Gateway Nodes (gateway1, gateway2)

- BGP Daemon (Quagga)

- MidoNet
  - Agent (Midolman)

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2. Basic Environment Configuration

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Networking Configuration

Important

All hostnames must be resolvable, either via DNS or locally.

This guide assumes that you follow the instructions in OpenStack Networking (neutron) of the OpenStack Documentation.

Repository Configuration

Configure necessary software repositories and update installed packages.

1. **Configure Ubuntu repositories**

   Edit the `/etc/apt/sources.list` file to contain the following:

```
# Ubuntu Main Archive
deb http://archive.ubuntu.com/ubuntu/ trusty main
deb http://security.ubuntu.com/ubuntu trusty-security main

# Ubuntu Universe Archive
deb http://archive.ubuntu.com/ubuntu/ trusty universe
deb http://security.ubuntu.com/ubuntu trusty-security universe
```

2. **Configure Ubuntu Cloud Archive repository**

   Create the `/etc/apt/sources.list.d/cloudarchive-kilo.list` file and edit it to contain the following:

```
# Ubuntu Cloud Archive
deb http://ubuntu-cloud.archive.canonical.com/ubuntu trusty-updates/kilo
 main
```

   Install the repository's key:

```
# apt-get update
# apt-get install ubuntu-cloud-keyring
```

3. **Configure DataStax repository**

   Create the `/etc/apt/sources.list.d/datastax.list` file and edit it to contain the following:

```
# DataStax (Apache Cassandra)
deb http://debian.datastax.com/community 2.0 main
```

   Download and install the repository's key:

```
# curl -L https://debian.datastax.com/debian/repo_key | apt-key add -
```

4. **Configure Java 8 repository**

   Since Ubuntu 14.04 does not provide a Java 8 runtime environment in its repositories, we are going to use the Launchpad PPA for OpenJDK.

   Create the `/etc/apt/sources.list.d/openjdk-8.list` file and edit it to contain the following:

```
# OpenJDK 8
deb http://ppa.launchpad.net/openjdk-r/ppa/ubuntu trusty main
```

   Download and install the repository's key:

```
# apt-key adv --keyserver hkp://keyserver.ubuntu.com:80 --recv-keys
 0x86F44E2A
```

5. **Configure MidoNet repositories**

   Create the `/etc/apt/sources.list.d/midonet.list` file and edit it to contain the following:

```
# MidoNet
deb http://builds.midonet.org/midonet-5 stable main

# MidoNet OpenStack Integration
deb http://builds.midonet.org/openstack-kilo stable main

# MidoNet 3rd Party Tools and Libraries
deb http://builds.midonet.org/misc stable main
```

   Download and install the repositories' key:

```
# curl -L http://builds.midonet.org/midorepo.key | apt-key add -
```

6. **Install available updates**

```
# apt-get update
# apt-get dist-upgrade
```

7. **If necessary, reboot the system**

```
# reboot
```


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3. OpenStack Installation

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Important

Follow the OpenStack Kilo Installation Guide for Ubuntu 14.04 (LTS) , but **note the following differences**.

Identity Service (Keystone)

Important

Follow the OpenStack documentation’s Chapter 3. Add the Identity service instructions, but **note the following differences and additions**.

1. **Verify operation**

   Do **not** apply step '1. For security reasons, disable the temporary authentication token mechanism'.

   The MidoNet API uses the Keystone admin token for authentication purposes, therefore **admin_token_auth** has to be kept in the corresponding configuration sections.

2. **Create MidoNet API Service**

   As Keystone admin, execute the following command:

```
$ openstack service create --name midonet --description "MidoNet API
 Service" midonet
```

3. **Create MidoNet Administrative User**

   As Keystone admin, execute the following commands:

```
$ openstack user create --password-prompt midonet
$ openstack role add --project service --user midonet admin
```

Compute Services (Nova)

Important

Follow the OpenStack documentation’s Chapter 5. Add the Compute service instructions, but **note the following differences**.

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Controller Node

Note

Follow the OpenStack documentation's Install and configure controller node instructions as is.

Compute Node

Important

Follow the OpenStack documentation's Install and configure a compute node instructions, but **note the following additions**.

1. **Configure libvirt**

   Edit the /etc/libvirt/qemu.conf file to contain the following:

```
user = "root"
group = "root"

cgroup_device_acl = [
    "/dev/null", "/dev/full", "/dev/zero",
    "/dev/random", "/dev/urandom",
    "/dev/ptmx", "/dev/kvm", "/dev/kqemu",
    "/dev/rtc","/dev/hpet", "/dev/vfio/vfio",
    "/dev/net/tun"
]
```

2. **Restart the libvirt service**

```
# service libvirt-bin restart
```

1. **Install nova-rootwrap network filters**

```
# apt-get install nova-network
```

2. **Restart the Compute service**

```
# service nova-compute restart
```

Networking Services (Neutron)

Important

Follow the OpenStack documentation's Chapter 6. OpenStack Networking (neutron) instructions, but **note the following differences**.

Controller Node

Important

Follow the OpenStack documentation's Install and configure controller node instructions, but **note the following differences and additions**.

1. **To configure prerequisites**

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Apply as is.

2. **To install the Networking components**

   Do **not** apply.

   a. Instead, install the following packages:

```
# apt-get install neutron-server python-neutronclient python-neutron-
plugin-midonet
# apt-get purge neutron-plugin-ml2
```

3. **To configure the Networking server component**

   Do **not** apply step 'd. Enable the Modular Layer 2 (ML2) plug-in, router service, and overlapping IP addresses'.

   a. Instead, edit the `/etc/neutron/neutron.conf` file and add the following key to the `[DEFAULT]` section:

```
[DEFAULT]
...
core_plugin = neutron.plugins.midonet.plugin.MidonetPluginV2
```

Note

Make sure to not leave any space at the starting of lines in any configuration file (this applies to all configuration files).

4. **To configure the Modular Layer 2 (ML2) plug-in**

   Do **not** apply.

   Instead, perform the following steps.

   a. Create the directory for the MidoNet plugin:

```
mkdir /etc/neutron/plugins/midonet
```

   b. Create the `/etc/neutron/plugins/midonet/midonet.ini` file and edit it to contain the following:

```
[DATABASE]
sql_connection = mysql://neutron:NEUTRON_DBPASS@controller/neutron

[MIDONET]
# MidoNet API URL
midonet_uri = http://controller:8080/midonet-api
# MidoNet administrative user in Keystone
username = midonet
password = MIDONET_PASS
# MidoNet administrative user's tenant
project_id = service
```

   c. Edit the `/etc/default/neutron-server` file to contain the following:

```
NEUTRON_PLUGIN_CONFIG="/etc/neutron/plugins/midonet/midonet.ini"
```

5. **To configure Compute to use Networking**

   Apply as is.

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6. **Configure Load-Balancer-as-a-Service (LBaaS)**

   Additionally to the OpenStack Installation Guide, configure Load-Balancer-as-a-Service (LBaaS) as described in the section called "Configure Load-Balancer-as-a-Service (LBaaS)" [9].

7. **To finalize installation**

   Do **not** apply.

   Instead, perform the following steps.

   a. Populate the database:

   ```
   # su -s /bin/sh -c "neutron-db-manage --config-file /etc/neutron/
   neutron.conf --config-file /etc/neutron/plugins/midonet/midonet.ini
    upgrade head" neutron
   ```

   b. Restart the Compute service:

   ```
   # service nova-api restart
   ```

   c. Restart the Networking service:

   ```
   # service neutron-server restart
   ```

Configure Load-Balancer-as-a-Service (LBaaS)

Important

Follow the OpenStack documentation's Configure Load-Balancer-as-a-Service (LBaaS) instructions, but **note the following differences**.

1. **Install the agent**

   Do **not** apply.

   Instead install the package python-neutron-lbaas

   ```
   # yum install python-neutron-lbaas
   ```

2. **Enable the HAProxy plug-in by using the service_provider option in the /etc/neutron/neutron.conf file:**

   Do **not** apply.

   Instead, set `service_provider` as follows:

   ```
   [service_providers]
   service_provider = LOADBALANCER:Midonet:midonet.neutron.services.
   loadbalancer.driver.MidonetLoadbalancerDriver:default
   ```

3. **Enable the load-balancing plug-in by using the service_plugins option in the /etc/neutron/neutron.conf file:**

   Apply as is.

4. **Enable the HAProxy load balancer in the /etc/neutron/lbaas_agent.ini file:**

   Do **not** apply.

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5. **Select the required driver in the /etc/neutron/lbaas_agent.ini file:**

   Do **not** apply.

6. **Create the required tables in the database:**

   Do **not** apply.

7. **Apply the settings by restarting the neutron-server and neutron-lbaas-agent services.**

   Do **not** apply.

8. **Enable load balancing in the Project section of the dashboard.**

   Apply as is.

9. **To finalize installation**

   Finalize the installation as described in Neutron Controller Node Installation.

Compute Node

Important

Follow the OpenStack documentation's Install and configure compute node instructions, but **note the following differences**.

1. **To configure prerequisites**

   Do **not** apply.

2. **To install the Networking components**

   Do **not** apply.

3. **To configure the Networking common components**

   Do **not** apply.

4. **To configure the Modular Layer 2 (ML2) plug-in**

   Do **not** apply.

5. **To configure the Open vSwitch (OVS) service**

   Do **not** apply.

6. **To configure Compute to use Networking**

   Apply as is.

7. **To finalize the installation**

   Do **not** apply.

   a. Instead, restart the following service:

   ```
   # service nova-compute restart
   ```

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4. MidoNet Installation

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NSDB Nodes

ZooKeeper Installation

1. **Install ZooKeeper packages**

```
# apt-get install openjdk-7-jre-headless
# apt-get install zookeeper zookeeperd zkdump
```

2. **Configure ZooKeeper**

   a. **Common Configuration**

   Edit the `/etc/zookeeper/conf/zoo.cfg` file to contain the following:

```
server.1=nsdb1:2888:3888
server.2=nsdb2:2888:3888
server.3=nsdb3:2888:3888
```

   Important

   For production deployments it is recommended to configure the storage of snapshots in a different disk than the commit log. This can be set by changing the parameter `dataDir` in `zoo.cfg` to a different disk.

   b. **Node-specific Configuration**

   i. **NSDB Node 1**

   Create the `/var/lib/zookeeper/myid` file and edit it to contain the host's ID:

```
# echo 1 > /var/lib/zookeeper/myid
```

   ii. **NSDB Node 2**

   Create the `/var/lib/zookeeper/myid` file and edit it to contain the host's ID:

```
# echo 2 > /var/lib/zookeeper/myid
```

   iii. **NSDB Node 3**

   Create the `/var/lib/zookeeper/myid` file and edit it to contain the host's ID:

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```
# echo 3 > /var/lib/zookeeper/myid
```

3. **Restart ZooKeeper**

```
# service zookeeper restart
```

4. **Verify ZooKeeper Operation**

   After installation of all nodes has been completed, verify that ZooKeeper is operating properly.

   A basic check can be done by executing the `ruok` (Are you ok?) command on all nodes. This will reply with `imok` (I am ok.) if the server is running in a non-error state:

```
$ echo ruok | nc 127.0.0.1 2181
imok
```

   More detailed information can be requested with the `stat` command, which lists statistics about performance and connected clients:

```
$ echo stat | nc 127.0.0.1 2181
Zookeeper version: 3.4.5--1, built on 06/10/2013 17:26 GMT
Clients:
 /127.0.0.1:34768[0](queued=0,recved=1,sent=0)
 /192.0.2.1:49703[1](queued=0,recved=1053,sent=1053)

Latency min/avg/max: 0/4/255
Received: 1055
Sent: 1054
Connections: 2
Outstanding: 0
Zxid: 0x260000013d
Mode: follower
Node count: 3647
```

Cassandra Installation

1. **Install Cassandra packages**

```
# apt-get install openjdk-7-jre-headless
# apt-get install dsc20
```

2. **Configure Cassandra**

   a. **Common Configuration**

   Edit the `/etc/cassandra/cassandra.yaml` file to contain the following:

```
# The name of the cluster.
cluster_name: 'midonet'

...

# Addresses of hosts that are deemed contact points.
seed_provider:
    - class_name: org.apache.cassandra.locator.SimpleSeedProvider
      parameters:
          - seeds: "nsdb1,nsdb2,nsdb3"
```

   b. **Node-specific Configuration**

   i. **NSDB Node 1**

Edit the `/etc/cassandra/cassandra.yaml` file to contain the following:

```
# Address to bind to and tell other Cassandra nodes to connect to.
listen_address: nsdb1

...

# The address to bind the Thrift RPC service.
rpc_address: nsdb1
```

ii. **NSDB Node 2**

Edit the `/etc/cassandra/cassandra.yaml` file to contain the following:

```
# Address to bind to and tell other Cassandra nodes to connect to.
listen_address: nsdb2

...

# The address to bind the Thrift RPC service.
rpc_address: nsdb2
```

iii. **NSDB Node 3**

Edit the `/etc/cassandra/cassandra.yaml` file to contain the following:

```
# Address to bind to and tell other Cassandra nodes to connect to.
listen_address: nsdb3

...

# The address to bind the Thrift RPC service.
rpc_address: nsdb3
```

3. **Clean existing data and restart Cassandra**

```
# service cassandra stop
# rm -rf /var/lib/cassandra/*
# service cassandra start
```

4. **Verify Cassandra Operation**

After installation of all nodes has been completed, verify that Cassandra is operating properly.

Important

If Cassandra fails to start and prints a "buffer overflow" error message in its log file, you may try associating `127.0.0.1` with the hostname in `etc/hosts` (so that `hostname -i` will show `127.0.0.1`). This may solve the Cassandra start problem.

A basic check can be done by executing the `nodetool status` command. This will reply with UN (Up / Normal) in the first column if the servers are running in a non-error state:

```
$ nodetool -host 127.0.0.1 status
[...]
Status=Up/Down
|/ State=Normal/Leaving/Joining/Moving
```


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```
--  Address    Load       Tokens  Owns   Host ID
       Rack
UN  192.0.2.1  123.45 KB  256     33.3%
 11111111-2222-3333-4444-555555555555  rack1
UN  192.0.2.2  234.56 KB  256     33.3%
 22222222-3333-4444-5555-666666666666  rack1
UN  192.0.2.3  345.67 KB  256     33.4%
 33333333-4444-5555-6666-777777777777  rack1
```

Controller Node

MidoNet API Installation

1. **Install MidoNet API package**

```
# apt-get install midonet-api
```

2. **Configure MidoNet API**

Edit the /usr/share/midonet-api/WEB-INF/web.xml file to contain the following:

```
<context-param>
    <param-name>rest_api-base_uri</param-name>
    <param-value>http://controller:8080/midonet-api</param-value>
</context-param>
```

```
<context-param>
    <param-name>keystone-service_host</param-name>
    <param-value>controller</param-value>
</context-param>
```

```
<context-param>
    <param-name>keystone-admin_token</param-name>
    <param-value>ADMIN_TOKEN</param-value>
</context-param>
```

```
<context-param>
    <param-name>zookeeper-zookeeper_hosts</param-name>
    <param-value>nsdb1:2181,nsdb2:2181,nsdb3:2181</param-value>
</context-param>
```

3. **Install Tomcat package**

```
# apt-get install tomcat7
```

4. **Configure Tomcat's Entropy Source**

Edit the /usr/share/tomcat7/bin/catalina.sh file to contain the following:

```
JAVA_OPTS="$JAVA_OPTS -Djava.security.egd=file:/dev/./urandom"
```

5. **Configure Tomcat's Maximum HTTP Header Size**

Edit the /etc/tomcat7/server.xml file and adjust the maximum header size for the HTTP connector:

```
<Connector port="8080" protocol="HTTP/1.1"
           connectionTimeout="20000"
           URIEncoding="UTF-8"
           redirectPort="8443"
           maxHttpHeaderSize="65536" />
```


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6. **Configure MidoNet API context**

   Create the `/etc/tomcat7/Catalina/localhost/midonet-api.xml` file and edit it to contain the following:

```
<Context
    path="/midonet-api"
    docBase="/usr/share/midonet-api"
    antiResourceLocking="false"
    privileged="true"
/>
```

7. **Restart Tomcat**

```
# service tomcat7 restart
```

MidoNet CLI Installation

1. **Install MidoNet CLI package**

```
# apt-get install python-midonetclient
```

2. **Configure MidoNet CLI**

   Create the `~/.midonetrc` file and edit it to contain the following:

```
[cli]
api_url = http://controller:8080/midonet-api
username = admin
password = ADMIN_PASS
project_id = admin
```

Midolman Installation

The *MidoNet Agent (Midolman)* has to be installed on all nodes where traffic enters or leaves the virtual topology, in this guide this are the **gateway1**, **gateway2** and **compute1** nodes.

1. **Install Midolman package**

```
# apt-get install openjdk-8-jre-headless
# apt-get install midolman
```

2. **Set up mn-conf**

   Edit `/etc/midolman/midolman.conf` to point mn-conf to the ZooKeeper cluster:

```
[zookeeper]
zookeeper_hosts = nsdb1:2181,nsdb2:2181,nsdb3:2181
```

3. **Configure access to the NSDB for all agents**

   This step needs to happen only once, it will set up access to the NSDB for all MidoNet Agent nodes.

   Run the following command to set the cloud-wide values for the ZooKeeper and Cassandra server addresses:

```
$ cat << EOF | mn-conf set -t default
zookeeper {
```

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```
    zookeeper_hosts = "nsdb1:2181,nsdb2:2181,nsdb3:2181"
}

cassandra {
    servers = "nsdb1,nsdb2,nsdb3"
}
EOF
```

Run the following command to set the Cassandra replication factor:

```
$ echo "cassandra.replication_factor : 3" | mn-conf set -t default
```

4. **Configure resource usage**

   Run these steps on **each agent host** in order to configure resource usage.

   ### Important

   For production environments the **large** templates are strongly recommended.

   a. **Midolman resource template**

      Run the following command to configure the Midolman resource template:

      ```
      $ mn-conf template-set -h local -t TEMPLATE_NAME
      ```

      Replace **TEMPLATE_NAME** with one of the following templates:

      ```
      agent-compute-large
      agent-compute-medium
      agent-gateway-large
      agent-gateway-medium
      default
      ```

   b. **Java Virtual Machine (JVM) resource template**

      Replace the default `/etc/midolman/midolman-env.sh` file with one of the below to configure the JVM resource template:

      ```
      /etc/midolman/midolman-env.sh.compute.large
      /etc/midolman/midolman-env.sh.compute.medium
      /etc/midolman/midolman-env.sh.gateway.large
      /etc/midolman/midolman-env.sh.gateway.medium
      ```

5. **Configure MidoNet Metadata Proxy for all agents**

   This step needs to happen only once, it will set up MidoNet Metadata Proxy for all MidoNet Agent nodes.

   Run the following commands to set the cloud-wide values for the MidoNet Metadata Proxy:

   ```
   $ echo "agent.openstack.metadata.nova_metadata_url : \"http:/
   /nova_metadata_host:nova_metadata_port\"" | mn-conf set
   $ echo "agent.openstack.metadata.shared_secret : shared_secret" | mn-
   conf set
   $ echo "agent.openstack.metadata.enabled : true" | mn-conf set
   ```

   nova_metadata_host, **nova_metadata_port**, and **shared_secret** should be replaced with appropriate values. They need to match with the corresponding Nova Metadata API configuration.

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nova_metadata_host and **nova_metadata_port** specify the address on which Nova accepts Metadata API requests. **shared_secret** has to be the same as specified by the "metadata_proxy_shared_secret" field in the "neutron" section of nova.conf.

The Nova side of the configuration for the metadata service is same as when using Neutron Metadata Proxy. See the OpenStack documentation for details:

Cloud Administrator Guide: Configure Metadata

6. **Start Midolman**

```
# service midolman start
```

MidoNet Host Registration

1. **Launch MidoNet CLI**

```
$ midonet-cli
midonet>
```

2. **Create tunnel zone**

MidoNet supports the Virtual Extensible LAN (VXLAN) and Generic Routing Encapsulation (GRE) protocols to communicate to other hosts within a tunnel zone.

To use the VXLAN protocol, create the tunnel zone with type 'vxlan':

```
midonet> tunnel-zone create name tz type vxlan
tzone0
```

To use the GRE protocol, create the tunnel zone with type 'gre':

```
midonet> tunnel-zone create name tz type gre
tzone0
```

Important

Make sure to allow GRE/VXLAN traffic for all hosts that belong to the tunnel zone. For VXLAN MidoNet uses UDP port 6677 as default.

1. **Add hosts to tunnel zone**

```
midonet> list tunnel-zone
tzone tzone0 name tz type vxlan

midonet> list host
host host0 name controller alive true
host host1 name gateway1 alive true
host host2 name gateway2 alive true
host host3 name compute1 alive true

midonet> tunnel-zone tzone0 add member host host0
 address ip_address_of_host0
zone tzone0 host host0 address ip_address_of_host0

midonet> tunnel-zone tzone0 add member host host1
 address ip_address_of_host1
zone tzone0 host host1 address ip_address_of_host1
```

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```
midonet> tunnel-zone tzone0 add member host host2
 address ip_address_of_host2
zone tzone0 host host2 address ip_address_of_host2

midonet> tunnel-zone tzone0 add member host host3
 address ip_address_of_host3
zone tzone0 host host3 address ip_address_of_host3
```

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5. BGP Uplink Configuration

MidoNet utilizes the Border Gateway Protocol (BGP) for external connectivity.

For production deployments it is strongly recommended to use BGP due to it's scalability and redundancy.

For demo or POC environments, alternatively static routing can be used. See the Operations Guide for details.

The following instructions assume below sample environment:

- One floating IP network
  - *192.0.2.0/24*
- Two MidoNet gateway nodes
  - *gateway1*, connecting to *bgp1* via *eth1*
  - *gateway2*, connecting to *bgp2* via *eth1*
- Two remote BGP peers
  - *bgp1*, *198.51.100.1*, AS *64513*
  - *bgp2*, *203.0.113.1*, AS *64514*
- Corresponding MidoNet BGP peers
  - *198.51.100.2*, AS *64512*
  - *203.0.113.2*, AS *64512*

Follow these steps to configure the BGP uplinks.

1. Determine the Keystone `admin` tenant ID

   Use the `keystone` command to determine the Keystone `admin` tenant's ID:

```
$ keystone tenant-list
+----------------------------------+----------+---------+
|                id                |   name   | enabled |
+----------------------------------+----------+---------+
| 12345678901234567890123456789012 |  admin   |   True  |
+----------------------------------+----------+---------+
```

2. Launch the MidoNet CLI and find the MidoNet Provider Router

```
$ midonet-cli
midonet-cli>
```

   Because the MidoNet Provider Router is not associated with a tenant, the active tenant has to be cleared (`cleart`) first.

```
midonet-cli> cleart

midonet-cli> router list
router router0 name MidoNet Provider Router state up
router router1 name Tenant Router state up infilter chain0 outfilter
 chain1
```

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In this example the MidoNet Provider Router is **router0**.

3. Load the `admin` tenant

   Before continuing with further configuration, the `admin` tenant has to be set (`sett`). Use the ID you got from Keystone above.

```
midonet-cli> sett 12345678901234567890123456789012
tenant_id: 12345678901234567890123456789012
```

4. Create virtual ports for the BGP sessions

   For each remote BGP peer, create a port on the MidoNet Provider Router that is going to be used for BGP communication.

```
midonet> router router0 add port address 198.51.100.2 net
 198.51.100.0/30
router0:port0

midonet> router router0 add port address 203.0.113.2 net 203.0.113.0/30
router0:port1

midonet> router router0 port list
port port0 device router0 state up mac ac:ca:ba:11:11:11
 address 198.51.100.2 net 198.51.100.0/30
port port1 device router0 state up mac ac:ca:ba:22:22:22
 address 203.0.113.1 net 203.0.113.0/30
[...]
```

   In this example the created ports are **port0** and **port1**.

5. Configure BGP on the virtual ports

```
midonet> router router0 set asn 64512
midonet> router router0 add bgp-peer asn 64513 address 198.51.100.1
router0:peer0

midonet> router router0 list bgp-peer
peer peer0 asn 64513 address 198.51.100.1

midonet> router router0 add bgp-peer asn 64514 address 203.0.113.1
router0:peer1

midonet> router router0 list bgp-peer
peer peer0 asn 64513 address 198.51.100.1
peer peer1 asn 64514 address 203.0.113.1
```

6. Add routes to the remote BGP peers

   In order to be able to establish connections to the remote BGP peers, corresponding routes have to be added.

```
midonet> router router0 route add src 0.0.0.0/0 dst 198.51.100.0/30
 port router0:port0 type normal
router0:route0

midonet> router router0 route add src 0.0.0.0/0 dst 203.0.113.0/30
 port router0:port1 type normal
router0:route1
```

7. Advertise BGP routes

In order to provide external connectivity for hosted virtual machines, the floating IP network has to be advertised to the BGP peers.

```
midonet> router router0 add bgp-network net 192.0.2.0/24
router0:net0

midonet> router router0 list bgp-network
net net0 net 192.0.2.0/24
```

8. Bind virtual ports to physical network interfaces

   Bind the MidoNet Provider Router’s virtual ports to the physical network interfaces on the Gateway Nodes.

   ### Important

   Ensure that physical interfaces are in state UP and do not have an IP address assigned.

   a. List the MidoNet hosts and find the Gateway Nodes:

   ```
   midonet> host list
   host host0 name gateway1 alive true
   host host1 name gateway2 alive true
   [...]
   ```

   In this example the hosts are **host0** and **host1**.

   b. List the Gateway Nodes' physical interfaces:

   ```
   midonet> host host0 list interface
   [...]
   iface eth1 host_id host0 status 3 addresses [] mac 01:02:03:04:05:06
    mtu 1500 type Physical endpoint PHYSICAL
   [...]

   midonet> host host1 list interface
   [...]
   iface eth1 host_id host0 status 3 addresses [] mac 06:05:04:03:02:01
    mtu 1500 type Physical endpoint PHYSICAL
   [...]
   ```

   c. Bind the physical host interfaces to the MidoNet Provider Router’s virtual ports:

   ```
   midonet> host host0 add binding port router0:port0 interface eth1
   host host0 interface eth1 port router0:port0

   midonet> host host1 add binding port router0:port1 interface eth1
   host host1 interface eth1 port router0:port1
   ```

   d. Configure a stateful port group:

   ```
   midonet-cli> port-group create name uplink-spg stateful true
   pgroup0
   ```

   e. Add the ports to the port group:

   ```
   midonet> port-group pgroup0 add member port router0:port0
   port-group pgroup0 port router0:port0

   midonet> port-group pgroup0 add member port router0:port1
   port-group pgroup0 port router0:port1
   ```


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```
midonet> port-group pgroup0 list member
port-group pgroup0 port router0:port0
port-group pgroup0 port router0:port1
```

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6. Further Steps

MidoNet installation and integration into OpenStack is completed.

You can now continue with the creation of initial networks in Neutron.

Note

Consult the **Operations Guide** for further instructions on operating MidoNet.

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